



UGTOMS Seed-First Chrono Evidence

Lossless observed-video program, literal UGLUT2 traversal, classical 3D+time gates, and Grove/POCO execution strategy

PROFILE 0.3 - VERIFIED BASELINE + IMPLEMENTATION ROADMAP

Supplied fixture: sam_2353410928515192.mp4

Authoring date: 2026-08-30 | RTX 5070 Ti Laptop GPU 12 GB | POCO X7 Pro target

Verdict

The substrate now executes a custom exact observation program with no stored pixel map and no conventional video payload. The minimum traversal seed is eight bytes. Arbitrary camera novelty cannot be reconstructed from that seed unless the evidence exists elsewhere; exact novelty therefore remains part of the executable program. Physical 3D is still UNKNOWN until classical calibrated evidence passes promotion gates.

WHAT IS TRUE NOW

Executive verdict

STATUS	RESULT	BOUNDARY
VERIFIED	Custom seed-driven exact RGB24+PTS replay	229/229 source frames
VERIFIED	8-byte fixed-profile traversal seed	Addressing only
IMPLEMENTED	UGCODE24 q709 reversible codeword	227 selected frames
ACTIVE	Portable C++ and Grove/POCO runtime	Not yet device-proven
UNKNOWN	Metric 3D, hidden surfaces, complete human	No promotion from this clip

Implemented outcome: a fixed custom player can regenerate the complete raster traversal from the seed and one shared literal UGLUT2, decode exact novelty, invert UGCODE24, and restore all accepted RGB24 frames and timing. This is a seed-first substrate program, regardless of whether the engineering files retain codec-oriented names.

- New q709 output: 122,540,032 bytes; SHA-256 f1a87daa...564ccd5.
- Fresh-process source comparison: PASS for 229/229 RGB24 hashes and every half-open PTS interval.
- Address definition: one 8-byte root seed, a 128-byte self-verifying UGTRV1 recipe, and one 144-byte UGLUT2.
- No serialized pixel permutation, H.264, AV1, ZIP, MediaCodec bitstream, DINO feature payload, or generated hidden content.
- RTX authoring is exact but not real-time: 262.598 seconds for the 9.120-second fixture; peak CUDA allocation 682.031 MiB.

Byte scale (linear)

Lossy MP4 source	 12,132,305 B
Exact seed-program	 122,540,032 B
Decoded RGB24	 633,139,200 B

The 12.13 MB MP4 is lossy and is not an equal-quality compression baseline. The exact seed-program is 19.3544% of decoded RGB24 and 10.1003 times the source MP4.

EIGHT BYTES ARE REAL - AND LIMITED

Seed-first meaning and the hard information boundary

8 B root seed	128 B self-verifying recipe	144 B shared UGLUT2	122.54 MB exact observation program
------------------	--------------------------------	------------------------	--

The seed regenerates structure. Camera novelty remains explicit evidence.

If evidence is baked into the APK or fetched elsewhere, the 8-byte seed selects relocated data; it does not erase it.

The literal seed artifact **sam_2353410928515192.ugtoms-traversal-seed64** is exactly 8 bytes: hex 1867BAFA7C80C31F, SHA-256 a55cbc1b . . . 396e978a. The fixed profile supplies recipe seed 1, so only the 64-bit root is stored.

A fixed decoder maps each 64-bit seed to at most one output. It can therefore regenerate only data implied by its fixed program and that seed. UGLUT2 supplies quantized radius/direction; SplitMix64 supplies deterministic lineage; neither contains the observed colors.

A seed-only exact result is possible when the scene was originally produced by the same substrate grammar. For an external camera, any sample not implied by prior accepted state is exogenous novelty. Negative memory suppresses repeated facts; it does not turn unseen values into known values.

A custom APK can hide the evidence in its binary, fetch it from a server, or retain it in memory. In all three cases the eight-byte file is a selector and the evidence has merely moved. A plausible generated replacement would violate the requested non-hallucination rule.

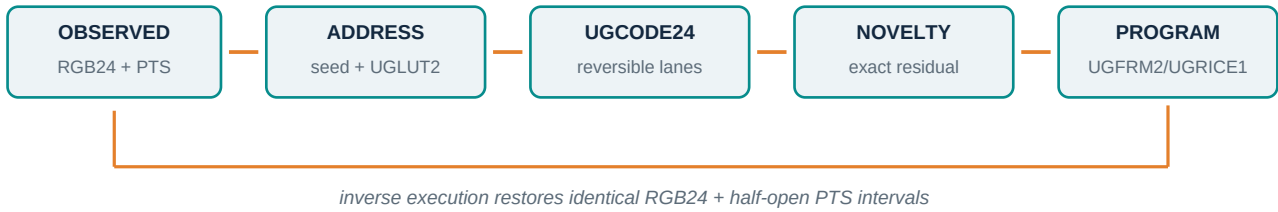
For a fixed deterministic decoder D and seed s:

```
structure = D_profile(seed, UGLUT2)
observation = structure + exact_novelty
seed_only exact <=> exact_novelty is empty
```

The supplied clip does not satisfy the final condition under the tested fixed grammar; its verified exact novelty dominates the file.

NO PICTURE-SIZED LUT

Literal substrate execution



```

a_i = UGTRV1(UGLUT2, root_seed, recipe_seed, i)
q_i = T(RGB[a_i])
e_i = (q_i - MED(q_left, q_up, q_upper_left)) mod 256
decode: entropy^-1 -> predictor^-1 -> T^-1 -> scatter[a_i]

```

UGLUT2: one 144-byte binary16 table of quantized sine, cosine and log-radius samples. It is a shared substrate dependency, not a raster lookup array.

UGTRV1: a new 128-byte chrono operator assembled from exact substrate primitives. It binds dimensions, two seeds, operator meaning, UGLUT2 SHA-256 and traversal SHA-256. It serializes no address list.

Traversal: doubled integer pixel centers; binary16 radius to Q16; direction to Q30; exact radial midpoint and cross-wedge classification; rho20/theta18 order; SplitMix64 lineage; Cartesian address as final tie-break.

UGCODE24: one reversible three-lane codeword per observed RGB8 sample. The regenerated address carries rho/theta/lineage, so those fields do not consume the same 24 color bits.

UGRICE1: canonical per-block selection among raw, signed modulo-256 Rice, and custom static 12-bit byte-rANS. All 9,847 blocks in the first full file selected byte-rANS.

THE USER'S THREE-LANE CORRECTION

UGCODE24 validation and exact file result

```

Cr = s8((R-G) mod 256)
Cb = s8((B-G) mod 256)
A  = floor((5*Cr + 2*Cb)/16)
Y  = (G+A) mod 256
store [Y,Cb,Cr]

```

$G = (Y-A) \bmod 256$; $R = (G+Cr) \bmod 256$; $B = (G+Cb) \bmod 256$

Validation	Result
Full-clip configuration matrix	29 configurations x 229 frames = 6,641 exact round trips; zero failures
All RGB24 inputs	16,777,216 codewords exhaustively inverted; PASS
Best all-intra stream	q709 [Y,Cb,Cr], 122,773,583 bytes
Authored mode selection	227 q709, 1 prior lift, 1 temporal
Authored file	122,540,032 bytes; 120,576 bytes below first exact file
Independent replay	Original MP4 re-decoded; all RGB24 hashes and PTS intervals exact

Exact layout under identical entropy	UGRICE1 stream bytes	Finding
q709 [Y,Cb,Cr]	122,773,583	Smallest measured all-intra
Prior q4 [Y,Cb,Cr]	122,881,215	Close second
Green differences	123,748,734	Larger
YCoCg-R	125,562,440	Larger
Lane bitplanes	316,524,893	Exact, but QR-like packing loses
24-bit grouped bitplanes	344,847,805	Exact, but substantially larger
Seed/address XOR	633,216,144	Destroys useful correlation

This validates the reversible codeword model and falsifies the assumption that QR-like bit rearrangement itself compresses this clip. The result is the smallest authored exact profile tested, not an absolute optimum.

REAL-TIME CAPTURE PLUS OFFLINE MINIMIZATION

RTX and POCO recorder/player path

Target	Verified capability	Next implementation
RTX 5070 Ti 12 GB	Exact CUDA gather/predict; 682 MiB peak	GPU-native traversal, parallel block entropy, seed/grammar search
POCO X7 Pro	API 36, 11.58 GB RAM, Mali-G720 MC7, 332 GB free	ARM64/NEON recorder/player in editable Grove scene
Camera2 back camera	FULL level, 1280x720 YUV at about 30 fps, REALTIME timestamps	Dense YUV authority, metadata/IMU binding, no implicit RGB
Grove engine	NativeActivity, GLES3, scene nodes, PTS staging reusable	Node-bound KCCH392 binding; no bootstrap/fullscreen special case

Capture authority: copy and normalize Camera2 YUV_420_888 planes immediately, hash them before prediction, and retain sensor PTS plus calibration/crop/exposure metadata. SurfaceTexture and device YUV-to-RGB conversion are presentation only.

Live path: the phone emits seed/recipe state and exact novelty into journaled chunks. A bounded queue may spill raw authoritative planes or stop explicitly under pressure; it must never silently drop a frame.

After-the-fact path: the RTX searches additional exact substrate recipes/transforms and rewrites the canonical smallest result found within the declared search budget. General absolute minimality is not computable; the report must say 'smallest measured among tested programs.'

Player: native entropy decode, inverse codeword/predictor, seed-regenerated traversal, two-slot publication on half-open PTS boundaries, and texture upload to an ordinary editable scene node.

Current fixture rate is about 13.44 MB/s, or about 48.4 GB/hour. The old APK-shell replacement projects to at least 129,464,915 bytes before native decoder, signing and alignment deltas. Neither real-time encode nor that APK size is yet device-verified.

GEOMETRY MUST EARN PROMOTION

Classical monocular 3D plus time

Outcome	Feasibility	Evidence requirement / current status
Exact 2D + time	HIGH - VERIFIED	Complete RGB24+PTS replay
Sparse projective support	CONDITIONAL	Tracks, non-degenerate motion, robust H/F gates
Relative Euclidean support	CONDITIONAL	Intrinsics, essential pose, parallax, cheirality; unknown scale
Metric geometry	NOT AVAILABLE	Measured calibration and scale anchor absent from fixture
Dense visible surface	CONDITIONAL	Repeated independent support plus bounded uncertainty
Hidden backs / closed topology	NOT IDENTIFIABLE	Remain UNKNOWN or explicit authored proxy
Complete human 4D	HIGH AMBIGUITY	Articulation, cloth, occlusion, identity and gauge branches

Implemented classical operators include GFTT, forward/backward KLT, seeded homography/fundamental estimation, symmetric-transfer and Sampson residuals, rank-two fundamental projection, essential-manifold projection, four pose branches, triangulation, cheirality, parallax, reprojection and conditioning gates.

Those operators currently produce guarded hypotheses only. The authored supplied-video file correctly stores empty geometry with UNBOUNDED_UNKNOWN; no mesh, voxel or human surface has been promoted.

All output primitives are same-time. Temporal continuity is entity/chart lineage, never a face, edge, voxel cell, splat or sheet connecting different source times.

Static pixels are observations too. 'Not detected as moving' is not proof of static support or free space. Every footprint remains a static candidate, independent-motion branch, occluder, unresolved sample or UNKNOWN until positive geometry supports a class.

BODY4D WITHOUT INVENTED ANATOMY

Human specialization and negative memory

May enter a human branch	Must remain UNKNOWN / PROXY_ONLY
Visible 2D joints and parts	Hidden anatomy and unobserved backs
Bounded/cyclic joint alternatives	Learned identity or body shape as truth
Same-time articulated transforms	Force, biomechanics, medical or biometric claims
Visibility and association alternatives	Closed topology inferred from silhouette
Explicitly authored proxy binding	Cloth/hair/skin continuity without evidence

Body4D, MoGe2, DA3, DINOv3, SAM3 and similar systems may propose masks, features, depth or pose only when their code, weights and configuration are bound. They cannot replace exact residuals or promote geometry.

DINOv3 is not required by the current seed program. A learned feature is not a bijection back to RGB and correlated model agreement is not independent physical measurement.

Negative memory stores accepted novelty: birth, tightening, split, chart discontinuity, owner handoff, retraction and checkpoint seal. Omission means no newly accepted fact. It never means disappearance, deletion, occlusion, empty space or mask complement.

Practical human outputs remain conditional: editable motion proxy, matchmove reference, partial visible support and downstream animation. They are not a recovered complete physical person.

USE CUSTOM STATE WHERE IT EARNS ITS COST

File formats and application ROI

Boundary	ROI decision
8-byte UGSEED64	Highest density for traversal selection; only by-reference, never standalone camera evidence
UGTC4D / seed program	Strong exact replay/provenance research; conventional compression advantage unproven
JSON/JSONL + JSON Schema	Highest immediate ROI for editable observations, hypotheses, novelty and receipts
Grove JSON / KC packs	High for editable scene ownership and hot native execution
PLY	First simple derivative for promoted sparse geometry; never canonical evidence
glTF 2.0.1 JSON / USDA	Presentation interchange only after geometry promotion; chrono GLB path not implemented
Alembic / OpenUSD / OpenVDB	Defer until validated time-slice meshes or sparse voxels exist
MP4	Source provenance and conventional playback baseline only; never custom frame payload
APK / AAR	Distribution only after native decoder and physical device evidence
PDF	Human review artifact, never machine authority

Highest chrono application ROI, in order:

- Deterministic author/verify and seed-program minimization CLI.
- Editable Grove evidence inspector with exact time scrubbing.
- Calibrated classical reconstruction workstation.
- Native POCO recorder/player with pull-back verification.
- Human-specific bounded articulated tooling after calibrated validation.

The broader repo still supports other high-ROI lanes, including the deterministic local pass planner and offline HTML delivery. These are evidence-adjusted engineering judgments, not revenue forecasts.

WHAT MAKES THE END STATE REAL

Execution roadmap and acceptance gates

Priority	Deliverable	Acceptance evidence
P0	Portable C++ reader	Python/C++ traversal, entropy, RGB and PTS parity; malformed files fail closed
P0	Native seed-program encoder	Scalar/NEON parity, journaled streaming, exact crash recovery
P0	Editable KCCH392 scene binding	Undoable node metadata; no bootstrap or hidden fullscreen object
P0	POCO recorder/player APK	No MP4/MediaCodec; installed artifact audit; 229-frame fixture replay
P0	Phone capture	Zero silent drops; plane hashes/PTS/metadata reproduce on phone, C++ and Python
P0	Performance	30 fps or explicit lossless spool mode; queues, PSS, thermals, energy measured
P1	Geometry record wiring	Real HYPOTHES/GEOMETRY sections; unsupported fields remain UNKNOWN
P1	Calibrated capture	Intrinsics, distortion, shutter, scale and independent support gates
P2	Human branch	Bounded visible joints/parts; no hidden-body promotion

Phone capture acceptance set: at least 10 seconds static/color/checkerboard and 10 seconds textured motion; monotonic unique sensor timestamps; capture count equals encoded ordinal count; exact pulled-file replay; no silent drops; and a kill-during-capture recovery test.

Compression kill gate: if seeded UGLUT2 ordering does not beat the same exact predictor/entropy under Cartesian, Morton, Hilbert or unseeded polar ordering, retain the format for execution/provenance but drop compression-superiority claims.

Geometry kill gate: if calibration, parallax, condition and independent-support gates do not contract uncertainty, materialize only observations/proxies and keep physical geometry UNKNOWN.

NO HALLUCINATED COMPLETION

Hard nonclaims and local evidence register

Claim boundary	Required wording
Lossless	Exact relative to declared PyAV RGB24+PTS decode, not original MP4 bits or photons
Seed	Regenerates traversal/program state; arbitrary camera novelty is retained elsewhere
3D	Exact pixels do not imply depth; projective support does not imply metric XYZ
Unknown	UNKNOWN is not free space, deletion, occlusion or a hidden surface
Time	No primitive crosses source time; continuity is lineage
Learned systems	Proposal/context only; agreement is not independent measurement
Phone	Capability is measured; custom runtime remains unverified until the APK runs
Optimality	Smallest authored among tested profiles, never absolute/global optimum

Primary local evidence

- spec/UGTOMS_CHRONO_GEOMETRY_CODEC_0_1.md
- src/ugts_kc3/chrono_substrate.py; chrono_seed.py; chrono_prediction.py
- src/ugts_kc3/chrono_entropy.py; chrono_codec.py; chrono_container.py
- src/ugts_kc3/chrono_compile.py; chrono_geometry.py; cli.py
- tests/test_chrono_seed.py; test_chrono_substrate.py; test_chrono_prediction.py
- tests/test_chrono_entropy.py; test_chrono_codec.py; test_chrono_container.py
- Parent sam_2353410928515192.ugtoms-seed-program-lossless-q709.ugtc4d
- Parent sam_2353410928515192.ugtoms-seed-program-lossless-q709.receipt.json
- Parent sam_2353410928515192.ugtoms-traversal-seed64
- Parent TODO.md and SUBSTRATE ROI STRATEGY.pdf

Final engineering position: execute every deterministic relation through the substrate, search hard for a compact exact seed program, retain only irreducible novelty, and never confuse omitted evidence with reconstructed physical truth.